Lithium-Ion Battery Recycling Equipment Guide

Comprehensive Machinery, Brands, and Technical Specifications for Black Mass Production

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1. Introduction

This document provides a comprehensive overview of the equipment required for lithium-ion battery recycling, specifically for black mass production through mechanical recycling processes. The information presented herein represents a synthesis of research into commercial lithium-ion battery recycling operations worldwide, with recommendations prioritized based on equipment that has demonstrated consistent performance at industrial scale.

The mechanical recycling approach described in this document represents the most economical, fastest, and most proven technology pathway for black mass production. Unlike pyrometallurgical methods that require high-temperature furnaces exceeding 1,400 degrees Celsius, mechanical recycling operates at ambient temperatures using standard industrial equipment. This approach has been successfully deployed at commercial scale by major recyclers worldwide, including Li-Cycle, Redwood Materials, BASF, Umicore, and Fortum.

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2. Executive Summary: Recommended Equipment Stack

Based on comprehensive research of commercial lithium-ion battery recycling operations worldwide, the following equipment stack represents the most reliable, proven technology for black mass production. This recommendation prioritizes equipment that has demonstrated consistent performance at industrial scale, with multiple global suppliers ensuring technology redundancy.

Process Stage	Recommended Equipment	Key Suppliers	Why Recommended
Primary Shredder	Rotary Shear (VR type) - Inertizable	BHS-Sonthofen, MTB, Andritz	Gas-tight design, nitrogen-purged, proven at BASF/Li-Cycle scale

Process Stage	Recommended Equipment	Key Suppliers	Why Recommended
Secondary Crusher	Universal Shredder (NGU type) or Hammer Mill	BHS-Sonthofen, Stokkermill, Metso	Fine liberation of electrode materials, adjustable particle size
Screening	Vibrating Screen + Rotary Drum	Major industrial suppliers	Standard tech, multiple suppliers, reliable sizing
Separation	Magnetic + Eddy Current + Air Classification	Eriez, Steinert, Metso	Proven technology, high recovery rates
Safety Systems	Nitrogen Inertization + Fire Suppression	Generon, fire system vendors	Critical for safe operation, prevents thermal events

3. Top Recommended Manufacturer: BHS-Sonthofen (Germany)

BHS-Sonthofen has emerged as the industry leader for lithium-ion battery recycling shredding systems based on their extensive commercial deployments and proven technology. Their equipment is currently used by BASF at their black mass production facility, representing one of the largest industrial-scale deployments in the world. The company has specifically developed inertizable shredders that can operate safely in nitrogen atmospheres, addressing the critical safety requirements of lithium battery processing.

Key BHS-Sonthofen Equipment for Black Mass Production

Equipment	Model	Function	Capacity	Key Features
Rotary Shear	VR Series	Primary shredding of battery packs/ modules	Up to 10 t/h	Gas-tight design, nitrogen inertizable, low speed (60-80 rpm), minimal heat generation
Universal Shredder	NGU Series	Secondary crushing to liberate materials	Up to 8 t/ h	Adjustable particle size, nitrogen compatible, high throughput, robust construction
Impact Crusher	BHS Impact Mill	Fine milling for black mass liberation	Variable	Produces consistent fine particles, optimized for electrode material separation

Commercial Validation

BHS-Sonthofen equipment has been validated through multiple commercial deployments. BASF operates a BHS-supplied plant for mechanical reprocessing of lithium-ion batteries into black mass, representing a major industrial-scale validation. The company has supplied multiple large-scale plants globally, with their inertizable shredders weighing between 12-18 tons and featuring unique capabilities for processing hazardous materials safely in protective nitrogen atmospheres.

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4. Alternative Reliable Manufacturers

While BHS-Sonthofen is the top recommendation, technology redundancy is essential for long-term operational security. The following manufacturers offer proven, reliable alternatives with demonstrated commercial deployments in lithium-ion battery recycling applications.

4.1 MTB Recycling (France)

MTB Recycling has developed specialized battery recycling equipment with a strong focus on safety and inert atmosphere processing. Their systems feature ATEX-compliant instrumentation and SIL 2 safety level components including temperature, pressure, and oxygen sensors. MTB offers complete lines for lithium battery processing with proven capability to handle EV battery packs, consumer batteries, and production scrap. Their equipment is particularly noted for robust separation technology and compliance with European safety standards.

4.2 ANDRITZ Group (Austria/Germany) + Duesenfeld Partnership

ANDRITZ has partnered with Duesenfeld GmbH, a German pioneer in lithium-ion battery recycling, to commercialize an innovative low-temperature recycling process. The Duesenfeld process is notable for being the only recycling technology that does not require exhaust gas scrubbing, with electrolyte recovery as a key feature. This partnership combines ANDRITZ's industrial manufacturing capability with Duesenfeld's patented vacuum drying and mechanical separation technology. The process achieves 72% recycling rate in mechanical recycling alone, with potential for higher rates through hydrometallurgical processing.

4.3 Stokkermill (Italy)

Stokkermill offers complete lithium battery recycling plants with mechanical shredding and separation systems. Their equipment is designed for cost-effective operation and has been deployed in multiple European facilities. Stokkermill provides turnkey solutions ranging from small-scale pilot plants to full industrial installations. Their hammer mills

and separation systems are particularly effective for mixed battery feed streams and offer good flexibility for varying input compositions.

4.4 Metso (Finland)

Metso provides comprehensive battery recycling solutions including their Battery Black Mass Recycling Process. While Metso focuses more on the hydrometallurgical processing of black mass, they also supply crushing and separation equipment for the mechanical pre-processing stage. Metso has supplied solvent extraction technology to Li-Cycle for their Rochester, New York facility, demonstrating their integration with major recyclers. Their equipment is known for reliability and integration with downstream processing.

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5. Equipment Comparison: Shredder Types

Shredder Type	Best For	Pros	Cons	Recommendation
Rotary Shear (VR)	Primary shredding of large battery packs	Low speed = less heat/sparks; handles large pieces; gas-tight designs available	Higher initial cost; larger footprint	HIGHLY RECOMMENDED for primary shredding
Hammer Mill	Secondary crushing; fine liberation	High throughput; proven technology; lower cost; good particle control	Higher heat generation; more wear parts	RECOMMENDED for secondary stage
Four-Shaft Shredder	Single-pass fine shredding	Consistent particle size; integrated screening	More complex; higher maintenance; not ideal for N2 atmosphere	ACCEPTABLE for consumer batteries only
Impact Crusher	Fine milling of pre-shredded material	Excellent liberation; adjustable output	Requires pre-sizing; not for whole batteries	RECOMMENDED for final liberation

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6. Recommended Complete Equipment Stack

The following represents the recommended equipment stack for a lithium-ion battery recycling facility focused on black mass production, optimized for reliability, proven technology, and multiple supplier redundancy:

Stage	Primary Equipment	Alternative Equipment	Key Specifications
1. Discharge	Automated discharge stations with resistive load banks	Saltwater bath systems (for small consumer batteries)	Discharge to <1V/cell; thermal monitoring; 4-24 hr cycles
2. Pre-sorting	Manual sorting stations with safety interlocks	Automated vision-based sorting (higher cost)	PPE including arc flash protection; fume extraction
3. Primary Shred	BHS VR Rotary Shear (inertizable, N2 purged)	MTB or Andritz equivalent low-speed shredder	Gas-tight; O2 sensors; 60-80 rpm; 50-100mm output
4. Secondary Crush	BHS NGU Universal Shredder or Hammer Mill	Stokkermill hammer mill; Metso crusher	Output <20mm; adjustable screens; N2 compatible
5. Fine Milling	Impact mill or ball mill for liberation	Additional hammer mill pass	Target <5mm for optimal black mass separation
6. Screening	Vibrating screen with multiple deck sizes	Rotary drum screen	Multiple fractions; recirculation of oversize
7. Magnetic Sep.	Drum magnet or overhead magnet	Eriez or Steinert magnetic separators	Extract ferrous materials; protect downstream
8. Eddy Current	Eddy current separator for non-ferrous metals	Steinert or Eriez ECS units	Recover Cu and Al current collectors
9. Air Classification	Air classifier or density separator	Zigzag separator; sink-float system	Separate black mass from lighter fractions
10. Collection	Dust collection system with HEPA filtration	Baghouse collector	Sealed collection; sampling port; moisture control

7. Critical Safety Systems

The following safety systems are essential and must be integrated with any battery recycling equipment. These are not optional and failure to implement them properly creates unacceptable safety risks.

System	Purpose	Specifications	Suppliers
Nitrogen Inertization	Displace oxygen to prevent fire/explosion during shredding	O2 < 5% in process area; continuous monitoring; automatic purging	Generon (membrane N2 generators), Air Products
Fire Suppression	Automatic extinguishing of thermal events	Water mist or inert gas (CO2/N2); rapid detection; full coverage	Major fire system vendors; Siemens; Tyco
Gas Detection	Detect HF, HCN, CO, flammable gases	ATEX rated; SIL 2 safety level; continuous monitoring	Drager; Honeywell; MSA Safety
Gas Scrubbing	Neutralize toxic gas emissions	Acid gas scrubbers; activated carbon; HEPA final filtration	Various industrial air quality vendors
Explosion Venting	Controlled pressure relief	ATEX compliant; properly sized for enclosure volume	Integrated with shredder enclosure design

8. Key Industry Insights

What Major Recyclers Use

Li-Cycle: Uses mechanical shredding technology to produce black mass, followed by hydrometallurgical processing. They have received solvent extraction technology from Metso for their Rochester, NY facility. Their approach emphasizes high recovery rates through optimized mechanical liberation.

Redwood Materials: Founded by former Tesla CTO JB Straubel, Redwood operates large-scale battery recycling in Nevada. They recycle EV production scrap for Panasonic and supply recovered materials back for new battery production. Their process focuses on recovering lithium, nickel, and cobalt at battery-grade purity.

BASF: Operates a BHS-Sonthofen-supplied plant for mechanical reprocessing of lithium-ion batteries into black mass. This represents a major validation of BHS equipment at industrial scale with one of the world's largest chemical companies.

Technology Trends

Recent developments in battery recycling equipment have focused on enabling safe processing of batteries without prior discharge or disassembly. New systems from BHS-Sonthofen and other manufacturers can process the largest EV battery packs in a single system with appropriate inertization, significantly improving throughput. The industry is

moving toward fully integrated lines that combine discharge, shredding, separation, and black mass collection in continuous processes, reducing manual handling and associated safety risks.

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9. Final Recommendations

1. **For Maximum Reliability**: Select BHS-Sonthofen rotary shear (VR) and universal shredder (NGU) for the primary and secondary crushing stages. This equipment has been validated at BASF's commercial-scale facility and offers the most robust inertizable design available.
2. **For Technology Redundancy**: Establish relationships with at least two equipment suppliers. MTB Recycling and Andritz/Duesenfeld provide viable alternatives with different technical approaches, ensuring supply chain resilience.
3. **For Budget Optimization**: Consider Chinese manufacturers (such as EcoNiLi or Suny Group) for secondary equipment where certification requirements permit. However, maintain Western equipment for critical shredding stages where safety systems integration is essential.
4. **For Mixed Feedstock**: Ensure equipment can handle your specific battery mix. EV packs require larger primary shredders with pack handling capability. Consumer electronics batteries can use smaller, finer shredding systems.
5. **For Safety Compliance**: Budget approximately 20-30% of equipment costs for safety systems integration including nitrogen inertization, fire suppression, gas detection, and emergency systems. This is not optional for safe operation.

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10. Key Equipment Supplier Directory

Company	Country	Products	Website
BHS-Sonthofen	Germany	Inertizable shredders, complete battery recycling lines	www.bhs-sonthofen.com
MTB Recycling	France	Battery recycling lines, ATEX-compliant equipment	www.mtb-recycling.fr
ANDRITZ / Duesenfeld	Austria/ Germany	Low-temperature process, electrolyte recovery	www.andritz.com
Stokkermill	Italy	Cost-effective recycling plants, hammer mills	www.eu.stokkermill.com
Metso	Finland	Black mass processing, separation equipment	www.metso.com

Company	Country	Products	Website
Eriez	USA	Magnetic and eddy current separators	www.eriez.com
Steinert	Germany	Sorting and separation technology	www.steinertglobal.com
Generon	USA	Nitrogen generators for inertization	www.generon.com

11. Technology Redundancy and Supply Chain Benefits

A key advantage of the mechanical recycling approach is the technology redundancy built into the equipment selection. All required equipment—including shredders, screens, magnetic separators, and eddy current separators—is manufactured by multiple global suppliers. This eliminates single-source dependencies and ensures:

- **Competitive pricing**: Multiple suppliers drive competitive pricing
- **Supply chain reliability**: No single point of failure in the supply chain
- **Serviceability**: Well-established maintenance procedures and trained technicians
- **Scalability**: Equipment can be sized to meet various throughput requirements

12. Conclusion

Selecting the appropriate equipment for lithium-ion battery recycling facilities requires careful consideration of safety, efficiency, reliability, and supply chain factors. The equipment recommendations provided in this document represent the best available technology for mechanical black mass production, with proven track records at commercial-scale facilities worldwide.

The combination of rotary shear primary shredders (preferably BHS-Sonthofen VR Series), hammer mill or universal shredder secondary crushers, vibrating screening equipment, magnetic and eddy current separation systems, and comprehensive safety systems provides a robust and reliable processing train for lithium-ion battery recycling operations.

The mechanical recycling approach offers significant advantages over pyrometallurgical methods, including faster processing times (typically 4-8 hours from input to black mass), lower capital investment, and proven commercial deployment at scale by major industry players including BASF, Li-Cycle, and Redwood Materials.

Document prepared for lithium-ion battery recycling facility planning and equipment procurement purposes. This guide represents a synthesis of current industry best practices and commercial equipment deployments.